ModelIt!

Teacher's Guide

How Soap Actually Destroys Viruses

The Chemical Reaction That Saves Millions of Lives Every Year

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-PS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-PS1-4

Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students investigate substance mixing and its effects.
Disciplinary Core Idea	**PS1.B Chemical Reactions** — Substances interact and new substances may form.
Crosscutting Concept	**Cause and Effect** — Students identify cause and effect in chemical interactions.

CAST Testing Alignment

MULTIPLE CHOICE:

A student washes their hands with plenty of soap but only for 5 seconds. Why might this be less effective than washing for 20 seconds?

CONSTRUCTED RESPONSE:

Explain why soap is more effective than water alone for destroying viruses with lipid membranes.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Handwashing with soap is one of the most powerful public health interventions ever discovered. The chemistry explains why it works so well.

Soap Chemistry

Soap molecules are amphiphilic - one end is hydrophilic (loves water) and the other is lipophilic (loves fat). This lets them bridge between water and lipids.

Virus Structure

Many viruses, including influenza and coronavirus, have a lipid membrane - a fatty outer coating that protects them and allows them to infect cells.

The Destruction

When soap contacts a lipid membrane, the lipophilic tails wedge between the fat molecules, breaking the membrane apart. Without its membrane, the virus falls apart.

Semmelweis Story

In the 1840s, Dr. Semmelweis proved handwashing saved lives. The medical establishment rejected him. He died in an asylum while millions died of preventable infections.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify soap concentration, water, soap effectiveness, and virus survival as components.

E — Establish Relationships
Students establish that soap and water positively affect effectiveness, while effectiveness NEGATIVELY affects virus survival.

V — Visualize & Model
Students build the handwashing chemistry model with the key negative relationship.

E — Evaluate Outcomes
Students compare water-only, soap-only, and proper handwashing scenarios.

R — Revise & Extend
Students add scrubbing time to model why 20 seconds matters.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Soap doesn't just wash viruses away - it DESTROYS them."*

Watch For

Show molecule animation if available.

Slide 2: Learning Objectives

2 min

Say: *"Today we learn the chemistry of clean hands."*

Watch For

Define amphiphilic.

Slide 3: Big Question

2 min

Say: *"HOW does soap destroy viruses?"*

Watch For

Take initial ideas.

Slide 4: LEVER Framework

2 min

Say: *"We'll model molecular demolition."*

Watch For

Preview activities.

Slide 5: Activity 1: Components

8 min

Say: *"We want Virus Survival Rate as close to ZERO as possible."*

Watch For

Emphasize the goal.

Slide 6: Activity 2: Connections

8 min

Say: *"The NEGATIVE relationship is the whole point - effectiveness REDUCES survival."*

Watch For

Ensure understanding.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test water only, soap only, and proper handwashing."*

Watch For

Students run all three.

Slide 8: Discoveries

5 min

Say: *"Every second gives soap more time to find viruses."*

Watch For

Practice 20-second wash.

Slide 9: STEM Challenge

10 min

Say: *"Use your science knowledge to create a campaign."*

Watch For

Form groups for campaign design.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Soap Concentration, Water Present
- **INTERNAL:** Soap Effectiveness, Virus Survival Rate

Relationships

- **Soap Concentration to Soap Effectiveness:** POSITIVE (+) — More soap molecules available to attack virus membranes.
- **Water Present to Soap Effectiveness:** POSITIVE (+) — Water activates and spreads soap across all surfaces.
- **Soap Effectiveness to Virus Survival Rate:** NEGATIVE (-) — More effective soap = fewer viruses survive. This is the goal!

Key Discoveries

- Soap DESTROYS viruses by breaking apart their lipid membranes
- Water alone can't disrupt the virus's protective coating
- Soap needs water to spread and work effectively
- This is why 20 seconds of proper handwashing matters

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Soap just rinses germs away"	Soap DESTROYS many viruses by breaking apart their lipid membranes. They don't survive to be washed away.
"Water alone is enough"	Water can't break the chemical bonds in lipid membranes. Only amphiphilic molecules like soap can.
"Antibacterial soap is better for viruses"	Antibacterial chemicals target bacteria, not viruses. Regular soap is equally effective against viruses.